THE NOVA RESEARCH CURRICULUM

General Science and Social Studies

Grade 4 · Term 2

Student edition

General Science: SC-4.4 matter, SC-4.5 energy, SC-4.6 friction and simple machines. Social Studies: SS-4.4 land forms, climate and disasters; SS-4.5 river civilisations.

General Science · lesson notes

- **SC-4.4 (Weeks 1–4):** matter has mass and takes up space; solids, liquids, gases; melting, freezing, evaporation, condensation (cold glass of water).
- **SC-4.5 (Weeks 5–8):** forms of energy (heat, light, sound, electrical, chemical, movement); energy changes (a torch: chemical → electrical → light); saving energy at home.
- **SC-4.6 (Weeks 9–12):** friction slows moving things and acts against motion; fair test: slide a book on wood, carpet and sandpaper; simple machines review: lever, pulley, wheel and axle, inclined plane, wedge, screw.

Social Studies · lesson notes

- **SS-4.4 (Weeks 1–8):** mountains, plateaus, plains, deserts; weather vs climate; floods, earthquakes, droughts in Pakistan and how to stay safe; how people adapt (houses, crops, clothing).
- **SS-4.5 (Weeks 9–12):** civilisations grew along rivers: Mesopotamia (Tigris and Euphrates), Egypt (Nile), and the Indus Valley (Mohenjo-daro, Harappa) with planned streets, drains and seals.

▶ **Try it on NovaSchool** (open **Simulations** on your dashboard):

- **The water cycle** — The Sun heats the sea. Water evaporates, cools into clouds, and falls as rain. Make the Sun stronger and watch.
- **Friction: a fair test** — Push a book with the same force across three surfaces. Only the surface changes — that makes it a fair test.

Workbook pages

Page	Outcome	Activity
1	SC-4.4	Changes of state diagram.
2	SC-4.5	Energy-change chains for a fan, torch and stove.
3	SC-4.6	Friction fair-test table and bar chart.
4	SS-4.4	Land forms of Pakistan map; earthquake safety steps.
5	SS-4.5	Compare three river civilisations in a table.

Question bank

#	Outcome	Question	Your answer
1	SC-4.4	Water turning into vapour is ...	freezing / melting / evaporation / condensation
2	SC-4.4	Water drops on a cold glass come from condensation. True or false.	True / False
3	SC-4.5	In a torch, the battery stores ... energy.	chemical / sound / light / heat
4	SC-4.5	Give two ways to save electricity at home.	_____
5	SC-4.6	Friction acts ... the direction of motion.	against / with / above / without
6	SC-4.6	On which surface does a book slide furthest?	sandpaper / carpet / smooth wood / grass
7	SC-4.6	A ramp is an inclined plane. True or false.	True / False
8	SS-4.4	Weather is the condition of the air ...	over many years / on a day / in space / under ground
9	SS-4.4	During an earthquake indoors you should ...	run outside at once / drop, cover and hold on / use the lift / stand by a window
10	SS-4.4	The Thar is a desert in Pakistan. True or false.	True / False
11	SS-4.5	Mohenjo-daro belongs to the ... civilisation.	Egyptian / Indus Valley / Greek / Roman
12	SS-4.5	The Egyptian civilisation grew along the ...	Indus / Tigris / Nile / Ravi
13	SS-4.5	Why did early civilisations grow near rivers?	_____